

MyOperator Campaigns API



This document describes a new Campaign Public API for making Peer to peer and IVR based calls.

1. Overview

MyOperator campaigns can be used to make various types of outgoing calls enabling client to connect with thier customers in ways which suites thier business requirement.

2. Type of Campaigns

According to the use cases, there are 2 types of campaigns which can be used to connect 2 users or connect customers with IVR or department:

2.1. Peer to Peer

Peer to peer are the type of calls which can be used to connect 2 parties (mostly customer and agent), MyOperator receives the API request to connect customer number with agent passed in the request. MyOperator will initiate the call to both parties one by one and connects them.

2.2. IVR based

IVR based outgoing calls are designed for bulk dialing cases where customer is connected to any agent who is free in the department. MyOperator receives API requests with customer number and

IVR ID of the outgoing IVR, MyOperator checks if any agent is free in the department provided in the IVR, call will be queued until someone in the department is free to accept call.

3. Authentication Setup

Before using this API, you need to create an account on myoperator.co, after setting up the account you need to apply for campaigns feature, your account manager should guide you through the setup process. After enabling features required in your use case, you will be able to access campaigns section on MyOperator panel located at app.myoperator.co/outbound/campaigns. Follow the documentation for campaigns section to get used to the user interface (#todo documentation link).

After setup you must have following things:

3.1. Setup Parameters

Parameters required in every API request for Authentication

Parameter	Required	Description
x-api-key	True	Key for accessing API resource
Secret Token	True	Secret for accessing API, unique for each account
Public IVR ID	True	ID referencing a unique campaign, multiple IVRs can be used in an account
Company ID	True	Unique ID of an account

Region	False	When using Dedicated DID, choose a region for call
Group	False	When using Dedicated DID, choose group of a region
Caller_id	False	When using Dedicated DID, choose a dedicated number to make call
max_call_duration	False	Define the maximum period after which the call should end once the customer and agents are connected (in seconds)
call_hold	False	Define whether to retry calls by sending request on hold, If UDC is unavailable.

4. Peer to Peer API

Peer to peer API can accept parameters in 2 ways, `user dial` and `anonymous dial`. In `user dial`, `user_id` parameter is passed along with a `number`. In `Anonymous dial` two numbers in parameters (`number` and `number_2`) are directly passed with any `user_id` in the request. `Anonymous dial` only works when Anonymous feature is enabled for the account. To enable anonymous feature please contact your account manager.

4.1. User Dial HTTP Request

User Dial API request parameters

Shell

```
curl --request POST 'https://obd-api.myoperator.co/obd-api-v1' \
--header 'x-api-key: {{x-api-key}}' \
--header 'Content-Type: application/json' \
```

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```

--data-raw '{
    "company_id": "{{Company ID}}",
    "secret_token": "{{Secret Token}}",
    "type": "1",      #1 for peer to peer
    "user_id": "{{User ID}}",
    "number": "<country_code><number>", #e.g. +919212992129
    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", #e.g. abd9238dh21ss
    "max_call_duration": <maximum call duration>, #e.g. 60
    "region": <region of a call>,
    "caller_id": <caller id number of a call>, #e.g. 8068492950
    "group": <group of a dedicated number>,
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
}'

```

Python

```

import requests

import json

url = "https://obd-api.myoperator.co/obd-api-v1"

payload = {
    "company_id": "{{Company ID}}",
    "secret_token": "{{Secret Token}}",
    "type": "1",      #1 for peer to peer
    "user_id": "{{User ID}}",
    "number": "<country_code><number>", #e.g. +919212992129
    "public_ivr_id": "{{Public IVR ID}}",

```

```

    "reference_id": "<client request id>", #e.g. abd9238dh21ss
    "max_call_duration": "<maximum call duration>", #e.g. 60
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>", #e.g. 8068492950
    "group": "<group of a dedicated number>",
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
}

payload = json.dumps(payload)

headers = {
    'x-api-key': '{{x-api-key}}',
    'Content-Type': 'application/json'
}

response = requests.request("POST", url, headers=headers, data = payload)
print(response.text.encode('utf8'))

```

PHP

```

<?php

$curl = curl_init();

curl_setopt_array($curl, array(
    CURLOPT_URL => "https://obd-api.myoperator.co/obd-api-v1",
    CURLOPT_RETURNTRANSFER => true,
    CURLOPT_CUSTOMREQUEST => "POST",
    CURLOPT_POSTFIELDS => '{

```

```

"company_id": "{{Company ID}}",
"secret_token": "{{Secret Token}}",
"type": "1",    //1 for peer to peer
"user_id": "{{User ID}}",
"number": "<country_code><number>", //e.g. +919212992129
"public_ivr_id": "{{Public IVR ID}}",
"reference_id": "<client request id>", //e.g. abd9238dh21ss
"max_call_duration": "<maximum call duration>", //e.g. 60
"region": "<region of a call>",
"caller_id": "<caller id number of a call>", //e.g. 8068492950
"group": "<group of a dedicated number>",
"call_hold": <whether to retry, if UDC not available>, #e.g. true
}',
CURLOPT_HTTPHEADER => array(
    "x-api-key: {{x-api-key}}",
    "Content-Type: application/json"
),
));

$response = curl_exec($curl);

curl_close($curl);

echo $response;

```

Nodejs

```

var request = require('request');

var body = {

    "company_id": "{{Company ID}}",

```

```

    "secret_token": "{{Secret Token}}",
    "type": "1",    //1 for peer to peer
    "user_id": "{{User ID}}",
    "number": "<country_code><number>", //e.g. +919212992129
    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", //e.g. abd9238dh21ss
    "max_call_duration": "<maximum call duration>", #e.g. 60
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>",// #e.g. 8068492950
    "group": "<group of a dedicated number>",
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
};

var options = {
    'method': 'POST',
    'url': 'https://obd-api.myoperator.co/obd-api-v1',
    'headers': {
        'x-api-key': '{{x-api-key}}',
        'Content-Type': 'application/json'
    },
    body: JSON.stringify(body)
};

request(options, function (error, response) {
    if (error) throw new Error(error);

    console.log(response.body);
});

POST https://obd-api.myoperator.co/obd-api-v1

```

In User dial, by default User is called first, if user picks up the call then `number` will be dialed. To reverse this direction (i.e. dial customer first), you can change the `Dial First` setting in the campaign section.

- `type` parameter value is always 1 for peer to peer dial.
- `reference_id` parameter gives the ability to generate your own unique id, which will be sent alongwith the webhook response.
- `user_id` is required parameter which is the user/agent to be dialed
- `number` should contain a valid phone number with coutry code, e.g. +919212992129.
- `max_call_duration` includes an integer value indicating the duration, in seconds, after which the call should be terminated. Maximum call duration that can be set is 5400 seconds. Note: There could be a delay of +-5 seconds in ending the call.
- `call_hold` controls whether to retry the call by sending request on hold in case UDC is unavailable. If this param is not passed, the default operation is to retry the call.
 - `call_hold = false`: Do not retry the call if the channel/user is unavailable. The call will get canceled. The agent will not receive any event regarding this.
 - `call_hold = true`: Retry the call by sending request to hold if the channel/user is unavailable.

Response of the API is provided in API response section.

4.2. Anonymous User Dial HTTP Request

Anonymous Dial API request parameters

Shell

```
curl --request POST 'https://obd-api.myoperator.co/obd-api-v1' \  
--header 'x-api-key: {{x-api-key}}' \  
--header 'Content-Type: application/json' \  
--data-raw '{  
    "company_id": "{{Company ID}}",  
    "secret_token": "{{Secret Token}}",  
    "type": "1",      #1 for peer to peer
```

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```

    "number": "<country_code><number>",
    "number_2": "<country_code><number>", #e.g. +919212992129
    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", #e.g. abd9238dh21ss
    "max_call_duration": "<maximum call duration>", #e.g. 60
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>", #e.g. 8068492950
    "group": "<group of a dedicated number>",
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
}

```

Python

```

import requests

import json

url = "https://obd-api.myoperator.co/obd-api-v1"

payload = {
    "company_id": "{{Company ID}}",
    "secret_token": "{{Secret Token}}",
    "type": "1",      #1 for peer to peer
    "number": "<country_code><number>", #e.g. +919212992129
    "number_2": "<country_code><number>", #e.g. +919999999999
    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", #e.g. abd9238dh21ss
    "max_call_duration": "<maximum call duration>", #e.g. 60
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>", #e.g. 8068492950
}

```

```

        "group": "<group of a dedicated number>",
        "call_hold": <whether to retry, if UDC not available>, #e.g. true
    }

payload = json.dumps(payload)

headers = {
    'x-api-key': '{{x-api-key}}',
    'Content-Type': 'application/json'
}

response = requests.request("POST", url, headers=headers, data = payload)

print(response.text.encode('utf8'))

```

PHP

```

<?php

$curl = curl_init();

curl_setopt_array($curl, array(
    CURLOPT_URL => "https://obd-api.myoperator.co/obd-api-v1",
    CURLOPT_RETURNTRANSFER => true,
    CURLOPT_CUSTOMREQUEST => "POST",
    CURLOPT_POSTFIELDS => '{
        "company_id": "{{Company ID}}",
        "secret_token": "{{Secret Token}}",
        "type": "1",    //1 for peer to peer
        "number": "<country_code><number>", //e.g. +919212992129
    }'

```

```

"number_2": "<country_code><number>", //e.g. +919999999999
"public_ivr_id": "{{Public IVR ID}}",
"reference_id": "<client request id>", //e.g. abd9238dh21ss
"max_call_duration": "<maximum call duration>", //e.g. 60
"region": "<region of a call>",
"caller_id": "<caller id number of a call>", //e.g. 8068492950
"group": "<group of a dedicated number>",
"call_hold": <whether to retry, if UDC not available>, #e.g. true
}',
CURLOPT_HTTPHEADER => array(
    "x-api-key: {{x-api-key}}",
    "Content-Type: application/json"
),
));

$response = curl_exec($curl);

curl_close($curl);
echo $response;

```

Nodejs

```

var request = require('request');

var body = {

    "company_id": "{{Company ID}}",
    "secret_token": "{{Secret Token}}",
    "type": "1",    //1 for peer to peer
    "number": "<country_code><number>", //e.g. +919212992129
    "number_2": "<country_code><number>", //e.g. +919999999999

```

```

    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", //e.g. abd9238dh21ss
    "max_call_duration": "<maximum call duration>", //e.g. 60
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>", //e.g. 8068492950
    "group": "<group of a dedicated number>",
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
};

var options = {
    'method': 'POST',
    'url': 'https://obd-api.myoperator.co/obd-api-v1',
    'headers': {
        'x-api-key': '{{x-api-key}}',
        'Content-Type': 'application/json'
    },
    body: JSON.stringify(body)
};

request(options, function (error, response) {
    if (error) throw new Error(error);
    console.log(response.body);
});
POST https://obd-api.myoperator.co/obd-api-v1

```

In Anonymous dial `Dial first` setting will work from IVR settings in campaigns section to change the order of dialing (`number` first or `number_2` first). You can also, switch the values of parameters `number` and `number_2` to achieve same.

- `type` parameter value is always 1 for peer to peer dial.
- `number` should contain a valid phone number with country code, e.g. +919212992129.
- `number_2` should contain a valid phone number with country code, e.g. +919212992129.
- `max_call_duration` includes an integer value indicating the duration, in seconds, after which the call should be terminated. Note: There could be a delay of +-5 seconds in ending the call.

- `call_hold` controls whether to retry the call by sending request on hold in case UDC is unavailable. If this param is not passed, the default operation is to retry the call.
 - `call_hold = false`: Do not retry the call if the channel/user is unavailable. The call will get canceled. The agent will not receive any event regarding this.
 - `call_hold = true`: Retry the call by sending request to hold if the channel/user is unavailable.

5. IVR based API

IVR based is used to connect a `number` with an IVR `public_ivr_id`. IVR can be configured in many ways to satisfy the given requirement, most common cases are:

- Connect `number` to any free user in the department mapped to the IVR.
- Connect `number` to automated IVR and collect input.
- Connect `number` with IVR configured with input node (feature for dynamic IVR execution).

IVR can be configured according to the requirement in [MyOperator campaigns section](#).

5.1. IVR based HTTP Request

IVR Dial API request parameters

Shell

```
curl --request POST 'https://obd-api.myoperator.co/obd-api-v1' \
--header 'x-api-key: {{x-api-key}}' \
--header 'Content-Type: application/json' \
--data-raw '{
  "company_id": "{{Company ID}}",
  "secret_token": "{{Secret Token}}",
  "type": "2",
  "number": "<country_code><number>", #e.g. +919212992129
```

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```
"public_ivr_id": "{{Public IVR ID}}",
"reference_id": "<client request id>", #e.g. abd9238dh21ss
"region": "<region of a call>",
"caller_id": "<caller id number of a call>", #e.g. 8068492950
"group": "<group of a dedicated number>"
}'
```

Python

```
import requests
import json

url = "https://obd-api.myoperator.co/obd-api-v1"

payload = {
    "company_id": "{{Company ID}}",
    "secret_token": "{{Secret Token}}",
    "type": "2",      #2 forIVR
    "number": "<country_code><number>", #e.g. +919212992129
    "public_ivr_id": "{{Public IVR ID}}",
    "reference_id": "<client request id>", #e.g. abd9238dh21ss
    "region": "<region of a call>",
    "caller_id": "<caller id number of a call>", #e.g. 8068492950
    "group": "<group of a dedicated number>",
    "call_hold": <whether to retry, if UDC not available>, #e.g. true
}

payload = json.dumps(payload)
```

```

headers = {
    'x-api-key': '{{x-api-key}}',
    'Content-Type': 'application/json'
}

response = requests.request("POST", url, headers=headers, data = payload)

print(response.text.encode('utf8'))

```

PHP

```

<?php

$curl = curl_init();

curl_setopt_array($curl, array(
    CURLOPT_URL => "https://obd-api.myoperator.co/obd-api-v1",
    CURLOPT_RETURNTRANSFER => true,
    CURLOPT_CUSTOMREQUEST => "POST",
    CURLOPT_POSTFIELDS => '{
        "company_id": "{{Company ID}}",
        "secret_token": "{{Secret Token}}",
        "type": "2", //2 for IVR
        "number": "<country_code><number>", //e.g. +919212992129
        "public_ivr_id": "{{Public IVR ID}}",
        "reference_id": "<client request id>", //e.g. abd9238dh21ss
        "region": "<region of a call>",
        "caller_id": "<caller id number of a call>", //e.g. 8068492950
        "group": "<group of a dedicated number>",
        "call_hold": <whether to retry, if UDC not available>, #e.g. true
    }'

```

```

    }',
21
    CURLOPT_HTTPHEADER => array(
22
23         "x-api-key: {{x-api-key}}",
24
25         "Content-Type: application/json"
26
27     ),
28
29 ));
30
31
32 $response = curl_exec($curl);
33
34
35 curl_close($curl);
36
37 echo $response;

```

Nodejs

```

1
2 var request = require('request');
3
4 var body = {
5
6     "company_id": "{{Company ID}}",
7
8     "secret_token": "{{Secret Token}}",
9
10    "type": "2",    //2 for IVR
11
12    "number": "<country_code><number>", //e.g. +919212992129
13
14    "public_ivr_id": "{{Public IVR ID}}",
15
16    "reference_id": "<client request id>", //e.g. abd9238dh21ss
17
18    "region": "<region of a call>",
19
20    "caller_id": "<caller id number of a call>", //e.g. 8068492950
21
22    "group": "<group of a dedicated number>",
23
24    "call_hold": <whether to retry, if UDC not available>, #e.g. true
25
26 };
27
28 var options = {
29
30     'method': 'POST',
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
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```

```

    'url': 'https://obd-api.myoperator.co/obd-api-v1',
    'headers': {
        'x-api-key': '{{x-api-key}}',
        'Content-Type': 'application/json'
    },
    body: JSON.stringify(body)
};

request(options, function (error, response) {
    if (error) throw new Error(error);

    console.log(response.body);
});

```

In IVR based calls `type` parameter is always 2. Settings which worked in peer to peer like `dial first` will not work since we only one parameter as number is passed. `number` is required parameter which is should contain country code. `reference_id` parameter gives the ability to generate your own unique id, which will be sent alongwith the webhook response.

6. Dedicated DID

Filters

Campaigns API has 3 parameters which you can use to manage the dedicated DID's. Before dedicated DID's can be sent as a parameter, you have to purchase them from Dedicated DID section and configure them in campaign IVR. Please check Campaign documentation for more detail.

Dedicated DID parameters (region, group, caller_id) are called `Dedicated DID filters`. User can use dedicated DID in calling process in many ways:

6.1 Filters not passed in params

If no dedicated DID filters are passed as a parameter, then MyOperator will check for any Dedicated DID assigned to the IVR (which can be managed from campaign settings section). If no Dedicated DID is assigned, call will go from shared route (if shared route is enabled in advanced settings in campaign section).

6.2 Filters passed in params

If filters are passed, one thing to be noted is, not all 3 filters need to be passed as parameters. MyOperator can automatically choose a random dedicated DID, if filter have many dedicated DID available. For example, if you have purchased 8 DIDs in 2 regions (Delhi and Noida) in 2 groups in each region, 2 DID in each group, like below:

Region	Group	DID
Delhi	gr1	11609821
Delhi	gr1	11609822
Delhi	gr2	11709821
Delhi	gr2	11709822
Noida	gr3	122666661
Noida	gr3	122666662
Noida	gr4	122766661
Noida	gr4	122766662

Different cases and how process will work is provided below:

Case	Params	Process
Only Region	region: "Delhi"	MyOperator will select a random group from available groups (gr1 , gr2) in region and selects a random DID from selected group. So MyOperator can randomly call from any DID in Delhi region
Only Group	group: "gr1"	MyOperator will select a random DID from gr1
Only DID	caller_id: "011609822"	MyOperator will call using DID 011609822
All params	region:"Delhi", group:"gr1", caller_id:"011609822"	MyOperator checks the params with priority caller_id, group, region. If higher priority param is found then rest params will be ignored. So, in this case since caller_id is passed, MyOperator will use caller_id parameter directly, other params (group and region) will be ignored.

Important points:

- Please double check if caller_id passed is purchased and it is in correct format, If caller_id parameter is invalid, API will accept the request and will not be able to call using dedicated DID.
- Using multiple groups and regions, will ensure the uptime of your service. MyOperator will automatically shift the load to another group (detecting and removing group might take few minutes). It is recommended to pass only `region` parameter and let MyOperator autoselect group according to availability.
- If you want to use only one DID for calling and want to use other DIDs as failover, then send DID in `caller_id` parameter, if group if selected DID is down, MyOperator can still use other Dedicated DID of any region if `failover` Global setting is enabled.

7. API Response

Whenever a request is received by campaigns API, an acknowledgement is sent in response with below parameters.

Success response body parameters

```
{  
  "details": "Request accepted successfully",  
  "status": "success",  
  "code": "200",  
  "unique_id": "<unique_generated_request_id>",  
  "reference_id": "<optional param will be present if provided in request_body>"  
}
```

Success response body (call_hold=false)

```
{  
  "details": "This call might get cancelled due to unavailability of  
UDC (User/Department/Channels)!!",  
  "status": "success",  
  "code": "200",  
  "unique_id": "<unique_generated_request_id>",  
  "reference_id": "<optional param will be present if provided in request_body>"  
}
```

Description of parameters of this acknowledgment message are provided below:

Parameter	Description
details	parameter gives the detailed message of the status of request.
status	status of request, could be either <code>success</code> or <code>error</code>

code	http status code of the request, different error codes are defined in errors section
unique_id	this is a unique id generated by MyOperator for a request
reference_id	if <code>reference_id</code> is present in the API request, same will be returned

8. Error codes

When an API request is sent, it could failed due to various reason, `code` parameter in API response can help in understanding the issue.

Missing or invalid x-api-key

If x-api-key parameter in header is missing, you will get error like below:

```

{
    "message": "Forbidden",
    "type": "INVALID_API_KEY",
    "code": "403",
    "status": "error",
    "stage": "prod",
    "resourcePath": "/obd-api-v1",
    "details": ""
}

```

Error response message body parameters

For other error messages will have below format:

```

{
  "details": "{msg}",
  "status": "error",
  "code": "<code>",
  "unique_id": "<unique_generated_request_id>",
  "reference_id": "<optional param will be present if provided in request_body>"
}

```

The MyOperator API uses the following error codes:

Error Code	Meaning
400	Bad Request -- Your request is invalid. check for missing parameter, request method etc.
403	Forbidden -- may cause due to billing issue, too many requests for same number, or invalid IP etc.
404	Not Found -- The specified resource could not be found. Check requested URL.
500	Internal Error -- We had a problem with our server. Try again later after some time or generate support ticket.

Some common error messages are:

Error Code	Message	Description
400	Provide valid Number in string format	

400	Provide valid Agent Number in string format	
400	Provide valid type in string format	type parameter is invalid, please check documentation
400	Provide only one a Valid user_id or number_2 in string format	
400	Provide either a Valid user_id or number_2 in string format	
400	Provide valid secret_token in string format	
403	Your IP is unauthorized access denied	Requester IP is not in Valid IP list, contact support
403	Your anonymous feature is not enabled	contact support team
403	User_id or number_2(agent_number) not allowed for this type	
403	Kindly choose any other reference ID, it has been already taken. Reference ID remaining_ttl x seconds	duplicate reference ID is not allowed for a certain time, please contact support team to change setting
403	Your billing usage has been Expired.	Billing issue please contact support team
403	Maximum number of requests for this number have exceeded.	Call on same number is allowed only few times in a day, please contact support to change setting.

404	Provide valid public_ivr_id in string format	
500	We are unable to process your request due to incomplete company_details	Configuration Issue, please contact support team